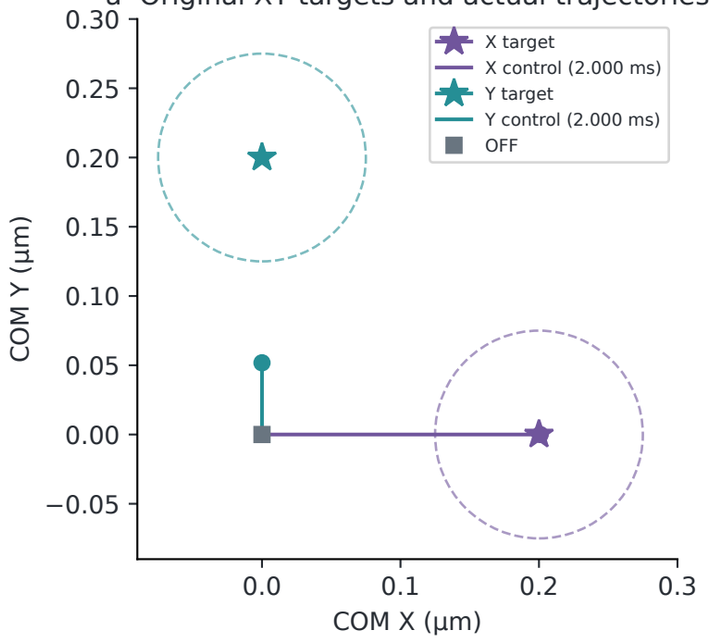
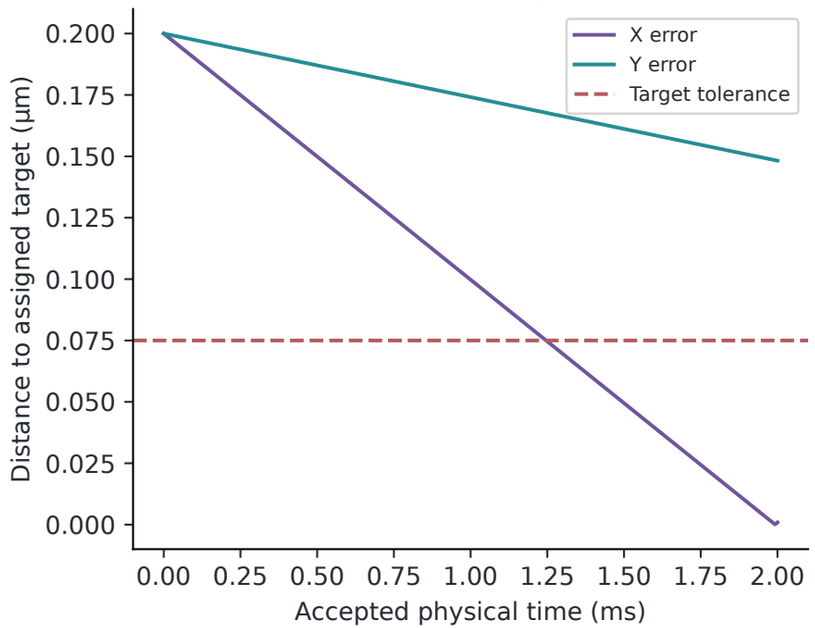
F10Q | Independent XY navigation with four PZTs and zero flow

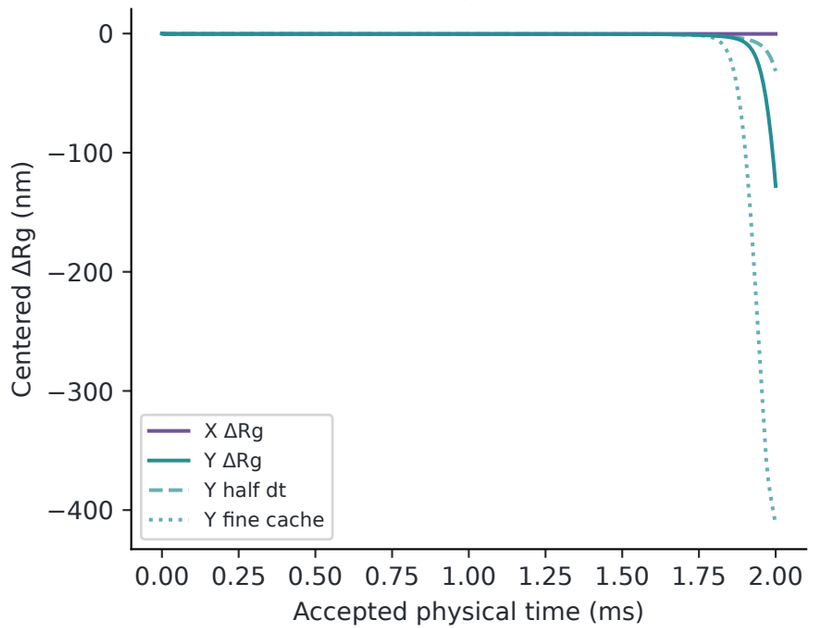
a Original XY targets and actual trajectories



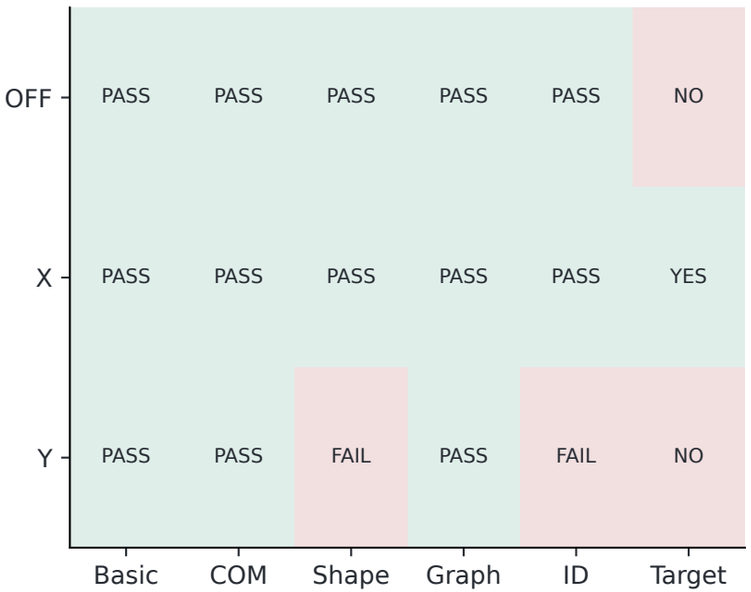
b Full-window target error



c Shape evolution, separate from translation



d Independent layered admission (2 ms)



e Same-model numerical differences

X dt: COM 0.000226 nm; ID 0.000893 nm
ΔRg 8.34e-07 nm; ΔQ/Q 3.9e-09

X cache: COM 0.0057 nm; ID 0.00575 nm
ΔRg 2.82e-07 nm; ΔQ/Q 1.28e-09

Y dt: COM 0.0001 nm; ID 396 nm
ΔRg 96.4 nm; ΔQ/Q 0.0404

Y cache: COM 0.00364 nm; ID 765 nm
ΔRg 307 nm; ΔQ/Q 0.122

f Scope, failures and actual computation

8 / 8 new controls: legal 0.5 ms screenings

Route A: 13.74 s; route B: 125.47 s

X: PASS; error 0.000896 µm

Y: PASS; error 0.148224 µm

5 / 10 / 30 ms: NOT_RUN

Exploratory group data; contact-model dependent

Experimental calibration: NOT_VERIFIED

TRAINING_READY = false

2.5 MHz · Homogeneous-water control proxy · PHYSICS_UNCALIBRATED · Shape/ID gates remain independent of COM admission.
EXPLORATORY · HOMOGENEOUS_WATER_CONTROL_PROXY · PHYSICS_UNCALIBRATED · training_ready=false